Solutions to Review Questions for Lecture 4

2025-2-6

1. `hist(coffee,freq=FALSE,breaks=seq(-0.5,50.5,1))`
2. The $25^{th}, 50^{th}$, and 75th percentiles.
3. Yes. A list of numbers where all of the numbers are all the same would have no whiskers.
4. False. Each of the first 3 blocks represent 10% of the data, and so the 30th percentile is 60, while the third block represents 30% of the data, so that the 60th percentile is 80.
5. True in this case. The 15th percentile will be halfway between 20 and 40, or 30. This is not, however, true in general.
6. The median will be the point that is 2/3 of the way from 60 to 80, since 60 is the 30th percentile and the bar from 60 to 80 represents 30% of the data. This value will be $60 + (2/3)(80 - 60) = 73.3$. The mean of each block will be in the middle since the values are evenly spread, with 10% of the values having mean 10, 10% mean 30, 10% mean 50, 30% mean 70, and 40% mean 90. Hence the mean will be $0.1 * 10 + 0.1 * 30 + 0.1 * 50 + 0.3 * 70 + 0.4 * 90 = 66$. This statement is false.
7. The calculation from 6 shows that the mean is greater than 50. The histogram would tip over to the right if we put our finger under 50, which means that the mean is greater than 50.
8. The mean is 66 and the differences from the mean are at most 66. Hence the standard deviation must be less than 66, which makes this statement false.
9. The median is 73.3 from problem 6.
10. The 25th percentile will be halfway between 40 and 60, or 50, and the 60th percentile will be 80. Hence the 75th percentile will be 15/40 of the way from 80 to 100, which is $80 + (15/40)(100) = 87.5$. This gives a value of $87.5 - 50 = 37.5$ for the IQR.
11. The 25th percentile will be roughly $-0.67 * SD$ while the 75th percentile will be roughly $0.67 * SD$ if the histogram follows the normal curve. Hence the $IQR = 0.67 * SD - (-0.67 * SD) = 1.34 * SD > SD$.
12. Markov's inequality gives $80/95 = 84\%$.
13. Let p be the proportion of the students who scored lower than 65 and let m_l be the mean for those students. Similarly, let m_u be the mean for the students who scored 65 or higher. We know that $80 = m_l * p + m_u * (1 - p)$. We also know that m_l is less than 65 and m_u is at most 100. Hence $80 = m_l * p + m_u * (1 - p) < 65p + 100 * (1 - p)$. Solving for p gives $p < 20/35 = 4/7$.
14. The score 95 is 3 standard deviations above the mean and the score 65 is 3 standard deviations below the mean. Hence Chebyshev's inequality gives a largest possible fraction of $1/3^2 = 1/9$.
15. $IQR > SD : (1, 2, 3, 4)$. The IQR will be $3 - 1 = 2$ and the $SD = \sqrt{(1 + 4 + 9 + 16)/4 - 2.5^2)} = 1.1$
$IQR < SD : (1, 2, 3, 100)$. The IQR will still be 2 and the $SD = \sqrt{(1 + 4 + 9 + 10000)/4 - (106/4)^2)} = 42.4$.
16. Markov's inequality gives $200000/1000000 = 20\%$. This is unrealistically large, as it assumes that the other houses will have \$0 price.

17. Let $\bar{X}$ be the mean of the list. Markov's inequality shows that the percentage of numbers that are $2\bar{X}$ or greater is at most $\bar{X}/2\bar{X} = 0.5$. Hence the percentage of numbers less than $2m$ is at least 50%, so that the median is at most $2m$.

18. The percentage of values that are more than $\sqrt{2}s$ above or below the mean is less than 1/2 by Chebyshev's inequality. This means that more than 1/2 of the values are greater than equal to $\bar{X} - s\sqrt{2}$ and more than 1/2 are less than or equal to $\bar{X} + s\sqrt{2}$. Hence $median(X) >= \bar{X} - s\sqrt{2}$ and $median(X) <= \bar{X} + s\sqrt{2}$, and so $|median(X) - \bar{X}| \leq s\sqrt{2}$.